Freezing Strategies in Transfer Learning for Skin Lesion Classification

Deep Learning Project Report S26

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1 Introduction and Background

Transfer learning has become a key method in deep learning, allowing models trained on large datasets to be efficiently adapted to new domains where data is often limited. A central question in this process is how much of the previously learned knowledge should be preserved, frozen, or fine-tuned when moving to a different task Yosinski et al. [2014].

This project examines different strategies for freezing layers in transfer learning applied to the classification of skin lesions, a task of clear clinical importance, where labeled data are scarce and accurate predictions can have direct diagnostic value Esteva et al. [2017]. By systematically testing various freezing configurations, we aim to better understand the balance between preserving general visual representations and adapting to task-specific features.

We take inspiration from an analogy in medical training: when a general practitioner specializes in dermatology, they keep their foundational understanding of medicine but develop new, focused expertise. Similarly, a pre-trained neural network retains basic visual patterns like edges and textures, while its higher-level layers often need fine-tuning to capture subtle distinctions between skin lesion types.

> *"When adapting a pre-trained model to a new task, how much of its previously learned knowledge should be retained?"*

2 Method

We use the HAM10000 dataset Tschandl et al. [2018], containing 10,015 dermoscopic images across 7 classes: melanoma (MEL), melanocytic nevi (NV), basal cell carcinoma (BCC), actinic keratosis (AKIEC), benign keratosis (BKL), dermatofibroma (DF), and vascular lesions (VASC). The dataset is significantly imbalanced, with NV dominating at $\sim$67% of samples.

2.1 Model Architecture

We use EfficientNet-B0 pre-trained on ImageNet as our primary backbone, due to its favorable accuracy–efficiency trade-off and widespread use in medical imaging benchmarks. ResNet18 is used for cross-architecture validation, allowing us to separate strategy effects from architecture effects.

Table 1: Three transfer-learning strategies evaluated on each backbone.

Strategy	Description	Expected Behaviour
S1 – From Scratch	All weights randomly initialised; full network trained on HAM10000	Weakest performance; lower bound baseline due to limited data
S2 – Full Freeze	All pre-trained weights frozen; only the classification head is trained	Fast training; may underfit if ImageNet features are insufficient
S3 – Gradual Unfreeze	Start with frozen backbone; progressively unfreeze layer blocks every few epochs	Best generalisation; balances retained knowledge with task-specific adaptation

We compare three freezing strategies, summarised in Table 1.

2.2 Training Setup

We split the dataset into training, validation, and test sets (80%, 10%, 10%) using stratified sampling to preserve class distribution across splits. Images are resized to 224×224 pixels and normalised using ImageNet mean and standard deviation. During training only, we apply random horizontal flips and rotations of up to $\pm 10^\circ$ for augmentation.

We use cross-entropy loss with class weights computed via `sklearn`'s `balanced` scheme to address the class imbalance, penalising mistakes on rare classes (DF, VASC, AKIEC) more than on common ones (NV). The Adam optimiser is used with an initial learning rate of 10^{-3}. For S3, the optimiser is rebuilt at 10^{-4} after each unfreezing event so newly trainable blocks are updated gently and pre-trained features are not disturbed Yosinski et al. [2014]. All strategies are trained for 10 epochs with a batch size of 32. The best checkpoint by validation accuracy is saved during training, and final metrics are reported on the held-out test set.

Limitation of training duration. Ten epochs is a short budget for S3; after each unfreezing event, newly trainable layers have limited time to specialise, so the S3 advantage over S2 is likely understated.

2.3 Implementation Details

The code is organised across several directories: `data/` for split CSVs and preprocessing, `models/` for model definitions (`model.py`, `resnet_model.py`), `train/` for training scripts, `evaluation/` for evaluation logic, `figures/` for generated plots, `results/` for saved outputs, and `configs/` for configuration files. Each strategy is launched via a dedicated training script located in `train/` (`train/train_s1_efficientnet.py`, `train/train_s2_efficientnet.py`, etc.). Frozen parameters are excluded from the optimiser using `filter(lambda p: p.requires_grad, model.parameters())`. For S3, EfficientNet blocks are unfrozen top-down via `model.features.children()`; for ResNet18, layers are unfrozen sequentially from `layer1` to `layer4`. The full codebase is available at `https://github.com/DaksaChella/skin-lesion-classification`.

3 Results

Table 2 reports the test-set metrics for the three freezing strategies on EfficientNet-B0. The ranking is monotone and unambiguous: S3 $>$ S2 $\gg$ S1.

Table 2: Test-set results on HAM10000 (EfficientNet-B0, 10 epochs). Higher is better for all metrics.

Strategy	Balanced Acc.	Macro F_1	Macro AUC
S1 – From Scratch	0.500	0.363	0.878
S2 – Full Freeze	0.618	0.461	0.912
S3 – Gradual Unfreeze	**0.655**	**0.486**	**0.924**

Table 3 reports test-set results for ResNet18, used for cross-architecture validation with S2 and S3 only. S1 is omitted as from-scratch performance is architecture-independent and already established in Table 2.

Table 3: Cross-architecture validation on HAM10000 (ResNet18, 10 epochs). Higher is better for all metrics.

Strategy	Balanced Acc.	Macro F_1	Macro AUC
S2 – Full Freeze	0.584	0.423	0.900
S3 – Gradual Unfreeze	**0.733**	**0.601**	**0.945**

Why the model performs as it does. S1 struggles across all rare classes, confirming that $\sim$8k training images are insufficient to train $\sim$5M parameters from scratch Tajbakhsh et al. [2016]. S2 lifts balanced accuracy by $+11.8$ points over S1 by reusing ImageNet edges/textures and learning a 7-way linear classification head. S3 adds a further $+3.7$ points on balanced accuracy and $+2.5$ on macro F_1 by progressively unfreezing feature blocks, allowing higher-level representations to specialise toward dermoscopic texture while a reduced learning rate (10^{-4}) protects the lower blocks Yosinski et al. [2014].

Note on S1 AUC. Despite near-random balanced accuracy (0.500), S1 achieves AUC 0.878 because the model learns to rank the dominant NV class confidently, inflating the AUC. This confirms that balanced accuracy and macro F_1 are more informative primary metrics for heavily imbalanced datasets.

Cross-architecture validation. On ResNet18, S3 clearly outperforms S2 ($+14.9$ points balanced accuracy, $+17.8$ macro F_1), confirming that gradual unfreezing is consistently beneficial across architectures. Notably, ResNet18 S3 (0.733) outperforms EfficientNet-B0 S3 (0.655), which is somewhat unexpected. We attribute this primarily to ResNet18's less compact feature representation providing more room for task-specific adaptation; the single random seed and small rare-class test samples (12–14) also mean these differences should be interpreted with caution.

Failure cases. The dominant error mode is rare-class$\rightarrow$NV confusion driven by the class prior ($\sim$2/3 of training samples are NV). Class-weighted loss mitigates but does not eliminate this; per-class F_1 for DF and VASC remains among the lowest across all strategies, reflecting the very small number of training examples for these classes.

Note on rare-class statistics. The test split contains only 12 DF and 14 VASC samples, making per-class scores for these classes statistically unreliable. This underlines why we report balanced accuracy and macro-averaged AUC as primary metrics, which are more robust to class imbalance than raw per-class accuracy.

4 Conclusion

What was achieved. A clean, reproducible comparison of three transfer-learning strategies on HAM10000 with two ImageNet backbones (EfficientNet-B0, ResNet18), with class-imbalance-aware training and evaluation on a held-out test set throughout.

Key findings. (1) Pre-trained features are essential — S1$\rightarrow$S2 alone closes most of the performance gap ($+11.8$ balanced accuracy points on EfficientNet-B0). (2) With 10 training epochs, S3 consistently outperforms S2 on both architectures: $+3.7$ points on EfficientNet-B0 and $+14.9$ on ResNet18 (test set). (3) Gradual unfreezing is particularly beneficial for architectures requiring stronger domain adaptation (ResNet18), but also advantageous for EfficientNet-B0 given sufficient training epochs.

Limitations. No learning-rate schedule, no hyperparameter sweep, single random seed, and no test-time augmentation. The very small number of DF and VASC test samples (12 and 14 respectively) makes per-class statistics for these classes unreliable and calls for multi-seed evaluation in future work.

Possible improvements. Longer training with cosine annealing; per-block discriminative learning rates rather than a single 10^{-4} post-unfreeze rate; oversampling rare classes (DF, VASC) in addition to weighted loss; stronger augmentation (colour jitter, dermoscopic-specific transforms); multi-seed evaluation to improve statistical reliability. Parameter-efficient adapters Hu et al. [2021] and lottery-ticket-style sparse fine-tuning Frankle and Carbin [2019], Morcos et al. [2019] are natural next steps.

References

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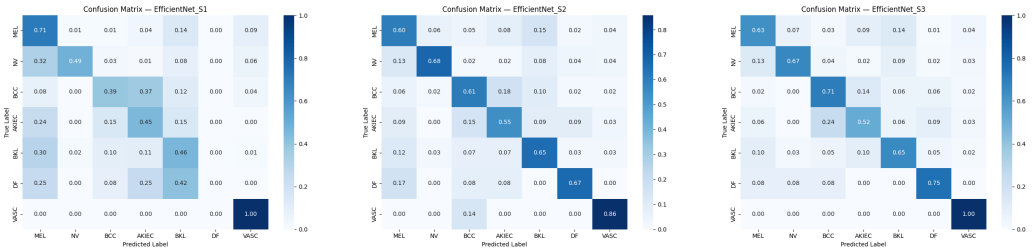
A Appendix

A.1 Confusion Matrices (EfficientNet-B0)

The diagonal improves consistently from S1 to S3. S1 shows widespread confusion across rare classes and DF is never correctly identified. S2 substantially improves the diagonal, and S3 further reduces rare-class errors with DF reaching 0.75 recall.

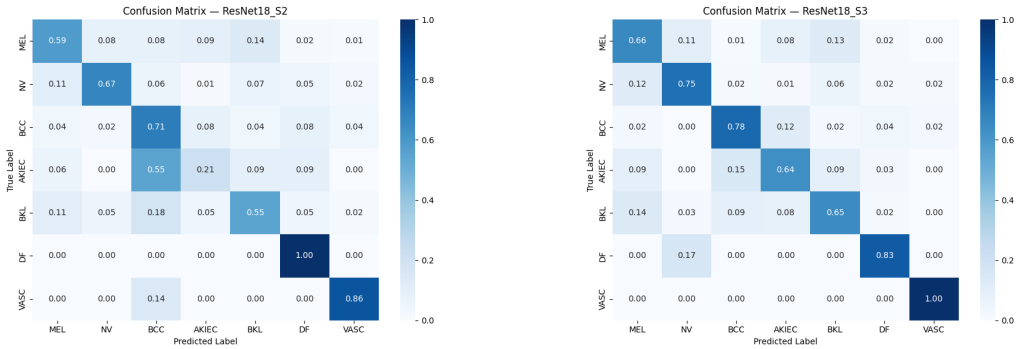


*S1 – From Scratch *S2 – Full Freeze *S3 – Gradual Unfreeze

Figure 1: Row-normalised confusion matrices on the HAM10000 validation split (EfficientNet-B0, 10 epochs).

A.2 Confusion Matrices (ResNet18)

ResNet18 S3 shows the strongest diagonal overall, particularly for BCC (0.78) and NV (0.75). The perfect DF recall in S2 (1.00) should be interpreted with caution given only 12 DF validation samples.

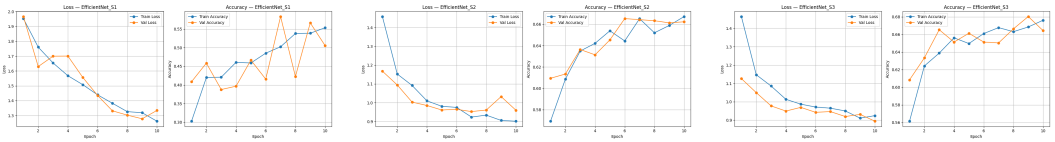


*S2 – Full Freeze *S3 – Gradual Unfreeze

Figure 2: Row-normalised confusion matrices on the HAM10000 validation split (ResNet18, 10 epochs).

A.3 Training Curves (EfficientNet-B0)

The S3 curve shows a steady improvement in both loss and accuracy throughout training, while S1 shows high variance in validation accuracy due to insufficient data for stable from-scratch training.

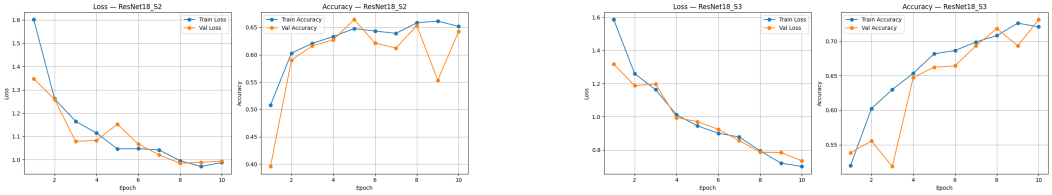


*S1 *S2 *S3

Figure 3: Per-strategy train/val loss and accuracy curves over 10 epochs (EfficientNet-B0).

A.4 Training Curves (ResNet18)

ResNet18 S3 shows the most consistent improvement, with both loss and accuracy converging smoothly over 10 epochs.



*S2 *S3

Figure 4: Per-strategy train/val loss and accuracy curves over 10 epochs (ResNet18).

A.5 Reproducibility

All experiments are launched from the project root. Splits are exported once via `python data/export_splits.py`. Strategies are reproduced with `train/train_s1_efficientnet.py`, `train/train_s2_efficientnet.py`, `train/train_s3_efficientnet.py`, `train/train_s2_resnet.py`, and `train/train_s3_resnet.py`. Evaluation and figures are generated with `evaluate.py`. Final test-set results are stored in `results/final_test_results.csv`.